Yes lets do this

Yes. I modeled the next step: replace the single photon mode by a **continuum of radiation modes**, and let the incoming external object's wavefunction change the strength of the same operator-valued Fx coupling.

The result is encouraging in the limited sense we wanted to test: the model produces ordinary exponential spontaneous emission in the isolated case, a much shorter lifetime when an external wavefunction overlaps the oscillator, and in both cases the emitted radiation remains centered at

$$\boxed{E_\gamma = \hbar\omega_0.}$$

Download the complete Python simulation package

Download the Python script

1. Initial state Ψ_1

Take the harmonic oscillator initially in its first excited state,

$$|1\rangle_{\rm osc},$$

the external object in an incoming wavefunction

$$|\chi_{\rm ext}\rangle,$$

and the radiation field in vacuum,

$$|0\rangle_\gamma.$$

Then

$$\boxed{|\Psi_1\rangle = |1\rangle_{\rm osc} \otimes |\chi_{\rm ext}\rangle \otimes |0\rangle_\gamma.} \tag{1}$$

This is initially a product state.

Your equation remains

$$\boxed{i\hbar\frac{\partial\Psi_1}{\partial t} + \frac{\hbar^2}{m}\frac{\partial^2\Psi_1}{\partial x^2} + i\hbar\frac{\partial\Psi_2}{\partial t} + \widehat{F}\widehat{x}\,\Psi_2 = 0.} \tag{2}$$

2. Replace the single photon by a continuum

Let

$$b_k^\dagger$$

create a photon in radiation mode k, with

$$E_k = \hbar\omega_k.$$

The oscillator position operator is

$$\boxed{\widehat{x} = x_0(\widehat{a} + \widehat{a}^\dagger), \qquad x_0 = \sqrt{\frac{\hbar}{2m\omega_0}}.} \tag{3}$$

Now choose the operator-valued force

$$\boxed{\widehat{F}(t) = \frac{\hbar}{x_0}\sum_k g_k\, f(t)\left(b_k + b_k^\dagger\right).} \tag{4}$$

The quantity

$$f(t)$$

will come from the external object's wavefunction.

Multiplying by $\widehat{x}$,

$$\widehat{F}(t)\widehat{x} = \hbar\sum_k g_k f(t)(b_k + b_k^\dagger)(a + a^\dagger).$$

Expanding,

$$\widehat{F}\widehat{x} = \hbar\sum_k g_k f(t)\left[ab_k^\dagger + a^\dagger b_k + ab_k + a^\dagger b_k^\dagger\right].$$

Near resonance, retaining the energy-exchange terms gives

$$\boxed{\widehat{F}\widehat{x} \simeq \hbar\sum_k g_k f(t)\left(ab_k^\dagger + a^\dagger b_k\right).} \tag{5}$$

This is the continuum version of the Fx term we discussed previously.

And the dimensions still work:

$$[g_k] = T^{-1},$$

so

$$[\hbar g_k] = E,$$

and

$$\boxed{[\widehat{F}] = \frac{E}{L} = \text{force},}$$

while

$$\boxed{[\widehat{F}\widehat{x}] = E.}$$

Therefore every term in your equation has dimensions

$$\boxed{\text{energy} \times \text{wavefunction}.}$$

3. Put the external object's wavefunction into the coupling

I represented the external object by a Gaussian wavepacket moving toward the oscillator.

The localized oscillator wavefunction and external wavefunction have a normalized overlap

$$\boxed{S(t) = \exp\left[-\frac{d(t)^2}{4(\sigma_{\text{osc}}^2 + \sigma_{\text{ext}}^2)}\right],} \tag{6}$$

where

$$d(t) = y_0 - vt.$$

Thus

$$S(t) \approx 0$$

when they are far apart, and

$$S(t) \to 1$$

when their centers coincide.

I used

$$\boxed{f(t) = 1 + \eta S(t).} \tag{7}$$

So in empty space,

$$S(t) = 0$$

and the oscillator has only its ordinary weak radiative coupling.

During an encounter,

$$S(t) > 0,$$

and the same Fx operator becomes stronger.

This was the numerical overlap:

External-wavefunction overlap and decay rate

4. The combined Ψ_2

Once oscillator and radiation become coupled, the combined state is

$$\boxed{|\Psi_2(t)\rangle = c_e(t)|1\rangle_{\rm osc}|0\rangle_\gamma + \sum_k c_k(t)|0\rangle_{\rm osc}|1_k\rangle_\gamma.} \tag{8}$$

I've suppressed the external-object factor here because in this first simulation its only role is to provide the overlap $S(t)$.

The amplitudes satisfy

$$i\dot{c}_e = \omega_0 c_e + \sum_k g_k f(t) c_k,$$

and

$$\boxed{i\dot{c}_k = \omega_k c_k + g_k f(t) c_e.} \tag{9}$$

So the system evolves continuously from

$$c_e(0) = 1, \qquad c_k(0) = 0$$

toward radiation states.

The oscillator population is

$$P_e(t) = |c_e(t)|^2,$$

while the total emitted-photon probability is

$$\boxed{P_\gamma(t) = \sum_k |c_k(t)|^2.} \tag{10}$$

And the simulation verifies

$$P_e + P_\gamma \simeq 1$$

to approximately 4×10^{-8}.

5. Why a continuum gives a lifetime instead of Rabi oscillations

With one radiation mode we obtained coherent exchange:

$$|1, 0\rangle \leftrightarrow |0, 1\rangle.$$

With hundreds of modes,

$$|0, 1_k\rangle,$$

the emitted amplitude disperses throughout the continuum and does not coherently return to the oscillator on the timescale studied.

In the Markov/Wigner–Weisskopf limit,

$$\boxed{\Gamma_0 = 2\pi J(\omega_0),} \tag{11}$$

where

$$J(\omega) = \rho(\omega)|g(\omega)|^2$$

is the radiation spectral density.

Without the external object,

$$\frac{dP_e}{dt} = -\Gamma_0 P_e,$$

giving

$$\boxed{P_e(t) = e^{-\Gamma_0 t}.} \tag{12}$$

Thus

$$\boxed{\tau_{\text{space}} = \frac{1}{\Gamma_0}.} \tag{13}$$

6. Your external wavefunction changes the lifetime

Since the interaction matrix element becomes

$$g_k \to g_k[1 + \eta S(t)],$$

and decay rates depend on the **square** of the matrix element,

$$\boxed{\Gamma(t) = \Gamma_0[1 + \eta S(t)]^2.} \tag{14}$$

Therefore

$$\boxed{P_e(t) = \exp\left[-\int_0^t \Gamma_0[1 + \eta S(t')]^2 dt'\right].} \tag{15}$$

This is a particularly simple realization of the idea you were proposing:

external wavefunction approaches

$\Downarrow$

$S(t) \uparrow$

$\Downarrow$

$\widehat{F}\widehat{x}$ coupling increases

$\Downarrow$

$\Gamma(t) \uparrow$

$$\Downarrow$$

$$\boxed{\text{shorter excited-state lifetime.}}$$

7. Numerical result

I used dimensionless units

$$\hbar = m = \omega_0 = 1,$$

with baseline decay rate

$$\Gamma_0 = 0.01.$$

So the expected isolated lifetime is

$$\tau_{\text{space}} \simeq 100.$$

The exact 501-mode continuum simulation gave

$$\boxed{\tau_{\text{space}}^{\text{exact}} = 100.51.}$$

The Markov prediction was

$$\boxed{\tau_{\text{space}}^{\text{Markov}} = 100.00.}$$

Very good agreement.

When the external wavefunction passed through the oscillator, using

$$\eta = 6,$$

the exact lifetime became

$$\boxed{\tau_{\text{encounter}}^{\text{exact}} = 22.54.}$$

The Markov result was

$$\boxed{\tau_{\text{encounter}}^{\text{Markov}} = 22.13.}$$

So in this illustrative model the external wavefunction shortened the lifetime by approximately

$$\boxed{\frac{100.51}{22.54} \simeq 4.46.}$$

You can see the complete comparison here:

Excited-state decay comparison

8. Where did the lost oscillator energy go?

Almost entirely into radiation.

At the end of the simulated interval, in the isolated case,

$$P_\gamma \simeq 0.6975,$$

because the simulation stopped after 120 time units while its lifetime is about 100.

For the external encounter,

$$\boxed{P_\gamma \simeq 0.9980.}$$

So essentially the entire oscillator excitation has become a photon.

Photon probability versus time

9. And the emitted energy really is $\hbar\omega_0$

This was perhaps the most important numerical check.

The conditional photon spectrum is

$$P(\omega_k) = \frac{|c_k|^2}{\sum_j |c_j|^2}.$$

Its mean frequency was

$$\langle \omega \rangle_{\text{space}} = 0.9999999981\,\omega_0,$$

and

$$\langle \omega \rangle_{\text{encounter}} = 0.9999999928\,\omega_0.$$

Thus

$$\boxed{\langle E_\gamma \rangle = \hbar \langle \omega \rangle \simeq \hbar \omega_0.} \tag{16}$$

The external object changes the **decay rate**, but not the transition energy.

Emitted photon spectrum

So your physical distinction works exactly as it should:

$$\boxed{\hbar\omega_0 = \text{energy quantum}}$$

while

$$\boxed{\widehat{F}\widehat{x} = \text{interaction controlling the transfer rate.}}$$

What this establishes—and what it doesn't

This simulation shows that an operator-valued term of your form

$$\boxed{\widehat{F}\widehat{x}}$$

can have the correct dimensions and operator structure to produce

$$|1\rangle_{\text{oscillator}} \to |0\rangle_{\text{oscillator}} + |1_\omega\rangle_{\text{radiation}},$$

and that coupling it to an incoming external wavefunction can naturally modify the lifetime while leaving

$$E_\gamma = \hbar\omega_0$$

unchanged.

However, the particular rule

$$f(t) = 1 + \eta S(t)$$

is an **additional interaction model**. Your bare equation does not yet tell us the value of η, the continuum couplings g_k, or that specific dependence on wavefunction overlap.

The next stronger version would keep the external object as a full quantum degree of freedom rather than reducing it to $S(t)$. Then we would write

$$\boxed{|\Psi_2\rangle = c_e|1\rangle|\chi_e\rangle|0\rangle_\gamma + \sum_k c_k|0\rangle|\chi_k\rangle|1_k\rangle,}$$

so oscillator, external object, and emitted radiation all become one genuinely entangled Ψ_2. That would let

us ask whether the lifetime enhancement can emerge from your equation from the **three wavefunctions themselves**, instead of being inserted through the phenomenological factor $\eta S(t)$.